Dorey's Resource Book

Topic 2 – Relative frequency and probability

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1 The probability scale

ANCHOR QUESTION

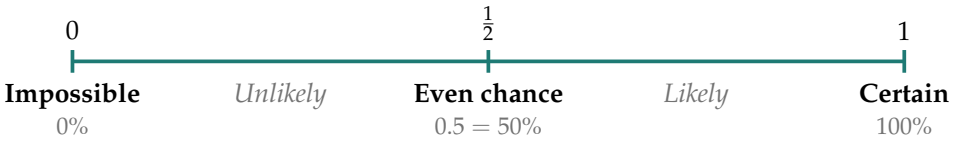
The probability that it rains tomorrow is 0.8. *Write this as a percentage and describe the chance in words.*

SYLLABUS MST-12-S1-07

- Recognise that probabilities range from 0 (impossible) to 1 (certain) and can be represented as fractions, decimals and percentages

NOTES

- **Probability** measures how likely an event is to happen.
- Every probability is between 0 and 1, including 0 and 1. As a percentage, it is between 0% and 100%.
- 0 means **impossible**, $\frac{1}{2}$ means **even chance** and 1 means **certain**.
- Between 0 and $\frac{1}{2}$ is **unlikely**. Between $\frac{1}{2}$ and 1 is **likely**.
- A probability can be written as a fraction, a decimal or a percentage: $\frac{1}{4} = 0.25 = 25\%$.
- To compare probabilities, write them all in the same form first.



WORKED EXAMPLE

The probability that a randomly chosen phone battery lasts more than two days is 0.25. *Write this as a percentage and describe the chance in words.*

$0.25 \times 100 = 25\%$ — decimal to percentage: $\times 100$
25% is less than 50% — compare with even chance
The chance is **unlikely**. — between 0 and $\frac{1}{2}$

Tier 1 · Decimal to percentage and words

1. Which of these numbers could be a probability?

$$0.4 \qquad 1.3 \qquad 75\% \qquad -0.1 \qquad 120\% \qquad \frac{2}{3}$$

2. Describe each probability in words.
(a) 0 (b) $\frac{1}{2}$ (c) 0.7 (d) 100%

3. The probability that a train arrives on time is 0.9. Write this as a percentage and describe the chance in words.

4. The probability that a new light globe is faulty is 0.02. Write this as a percentage and describe the chance in words.

Tier 2 · Starting from a fraction

5. The probability that a randomly chosen student walks to school is $\frac{2}{5}$. Write this as a decimal and as a percentage, and describe the chance in words.

6. The probability that a flight leaves on time is $\frac{17}{20}$. Write this as a decimal and as a percentage, and describe the chance in words.

7. The probability of winning a prize in a game is $\frac{1}{8}$. Write this as a decimal and as a percentage, and describe the chance in words.

Tier 3 · Starting from a percentage

8. A forecast gives a 60% chance of rain. Write this as a fraction in simplest form and as a decimal, and describe the chance in words.

9. In a raffle, 4% of the tickets win a prize. Write the probability that a ticket wins as a fraction in simplest form and as a decimal, and describe the chance in words.

10. A basketballer makes 72% of her free throws. Write the probability that she makes her next free throw as a fraction in simplest form and as a decimal, and describe the chance in words.

Tier 4 · Comparing and ordering

11. Order these probabilities from least likely to most likely: 0.3, $\frac{2}{5}$, 35%.

12. Order these probabilities from least likely to most likely: $\frac{7}{8}$, 0.85, 88%.

13. The probabilities of three teams winning a competition are: Hawks 0.45, Lions $\frac{9}{20}$, Owls 44%. Which team is least likely to win? Are any two teams equally likely to win?

Tier 5 · Judging from the situation

14. Use one of the words *impossible*, *unlikely*, *even chance*, *likely* or *certain* to describe each event.

(a) A fair coin lands on heads.

(b) A standard six-sided die is rolled and shows a 7.

(c) A randomly chosen month has at least 28 days.

15. One student from a class of 30 is chosen at random to be class captain. Which of the following best describes the probability that Ella, a student in the class, is chosen?

A. Certain

B. Even chance

C. Unlikely

D. Impossible

16. A bag holds 9 green counters and 1 red counter. A counter is taken out without looking. Describe, in words, the chance that it is green.

Conceptual

17. Explain why 1.4 cannot be a probability.

18. Sam says a probability of 0.08 is more likely than a probability of 0.5 because 8 is bigger than 5. Explain his mistake.

19. An event has an even chance of happening. Write its probability as a fraction, a decimal and a percentage.

20. A weather app gives a 0% chance of rain for tomorrow. Explain what this means, and why it might still rain.

21. The probability that a phone made at Factory A is faulty is 0.03.

(a) Write this probability as a percentage. (1 mark)

(b) Describe this chance in words. (1 mark)

(c) The probability that a phone made at Factory B is faulty is $\frac{1}{25}$. Which factory is more likely to make a faulty phone? Justify your answer. (2 marks)

Revision

22. A fruit punch is made by mixing juice and soda water in the ratio 3 : 2.

(a) What fraction of the punch is soda water? (1 mark)

(b) How many litres of juice are needed to make 4 L of punch? (2 marks)

(c) Juice costs \$3.60 per 2 L bottle. Find the cost of the juice for 4 L of punch. (2 marks)

23. Kira drives 342 km in 3 hours 48 minutes.

(a) Write 3 hours 48 minutes in hours, as a decimal. (1 mark)

(b) Find Kira's average speed in km/h. (1 mark)

(c) Convert this speed to m/s. (1 mark)

(d) Her car uses 7.5 L/100 km and petrol costs \$1.96 per litre. Find the cost of the fuel for the trip, to the nearest cent. (2 marks)

24. The number of goals scored by a netball team in eight games is shown.

4 7 7 8 9 10 12 15

(a) Find the mean. (1 mark)

(b) Find the median. (1 mark)

(c) Find the range. (1 mark)

(d) Find the interquartile range. (2 marks)

2 Sample spaces

ANCHOR QUESTION

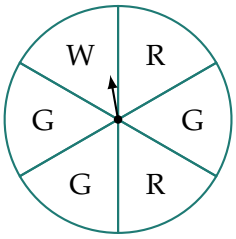
A standard six-sided die is rolled. List the sample space and state the number of outcomes.

SYLLABUS MST-12-S1-07

- Identify and define the sample space as the set of all possible outcomes of a chance experiment
- Determine the number of outcomes for an experiment

NOTES

- A **chance experiment** is an action with a result that cannot be predicted, such as tossing a coin or rolling a die.
- Each possible result is an **outcome**.
- The **sample space**, S, is the set of all possible outcomes, written in braces: tossing a coin gives $S = \{H, T\}$.
- List every possible outcome, each one **only once**.
- The number of outcomes is not always the number of sections: the spinner below has 6 sections but $S = \{R, G, W\}$, so 3 outcomes.



WORKED EXAMPLE

A spinner has 5 equal sections numbered 1 to 5. List the sample space and state the number of outcomes.

$S = \{1, 2, 3, 4, 5\}$ — every possible result, each once

5 outcomes — count the outcomes in S

Tier 1 · Listing the sample space

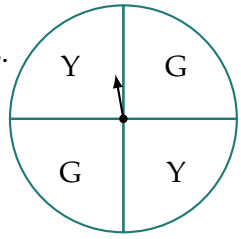
25. A coin is tossed. List the sample space and state the number of outcomes.

26. A spinner has 3 equal sections labelled A, B and C. List the sample space and state the number of outcomes.

27. A day of the week is chosen at random. List the sample space and state the number of outcomes.

Tier 2 · Repeated sections

28. List the sample space for this spinner and state the number of outcomes.



29. A bag holds 5 red, 3 blue and 2 white marbles. One marble is taken out without looking. List the sample space and state the number of outcomes.

30. A spinner has 6 equal sections numbered 1, 1, 2, 2, 2, 3. List the sample space and state the number of outcomes.

Tier 3 · Letters in a word

31. A letter is chosen at random from the word MISSISSIPPI. List the sample space and state the number of outcomes.

32. A letter is chosen at random from the word BALLOON. List the sample space and state the number of outcomes.

33. A letter is chosen at random from the word STATISTICS. List the sample space and state the number of outcomes.

Tier 4 · Counting without listing

34. A card is drawn from a standard deck of 52 playing cards. How many outcomes are in the sample space?

35. One student is chosen at random from a class of 28. How many outcomes are in the sample space?

36. Raffle tickets are numbered 100 to 499. One ticket is drawn. How many outcomes are in the sample space?

Tier 5 · Outcomes for an event

37. A standard die is rolled. List the outcomes that are greater than 4.

38. A number from 1 to 30 is chosen at random. List the outcomes that are multiples of 7.

39. A number from 1 to 20 is chosen at random. List the odd numbers greater than 12.

Conceptual

40. A spinner has 6 equal segments: 2 are red, 3 are green and 1 is white. Explain why its sample space is $S = \{R, G, W\}$ and not $S = \{R, R, G, G, G, W\}$.

41. Jo says the sample space for choosing a letter from the word BOOK is $\{B, O, O, K\}$, so there are 4 outcomes. Explain her mistake.

42. The sample space for a spinner is $S = \{1, 2, 3, 4, 5\}$. Could the spinner have 8 sections? Explain.

43. A soccer team plays a match. Write the sample space for the result of the match and state the number of outcomes.

44. A wheel has the numbers 1 to 30 marked on it. When the wheel is spun, each number is equally likely to win.

(a) How many outcomes are in the sample space? (1 mark)

(b) List all the even numbers on the wheel that are greater than 21. (1 mark)

(c) How many of the outcomes are NOT even numbers greater than 21? (1 mark)

Revision

45. The probability that a randomly chosen car at a service station is electric is $\frac{3}{40}$.

(a) Write this probability as a decimal. (1 mark)

(b) Write this probability as a percentage. (1 mark)

(c) Describe this chance in words. (1 mark)

(d) The probability that a randomly chosen car is a ute is 12%. Is a car more likely to be electric or a ute? Justify your answer. (2 marks)

46. Mia earns \$28.40 per hour for a 38-hour week. Overtime is paid at time-and-a-half.

(a) Find Mia's pay for 38 normal hours. (1 mark)

(b) Find Mia's overtime rate of pay. (1 mark)

(c) One week, Mia works 38 normal hours and 4 hours of overtime. Find her total pay for the week. (2 marks)

(d) Mia works the same hours the next week. Find her pay for the fortnight. (1 mark)

47. Rice is sold in a 1 kg bag for \$3.20 or a 5 kg bag for \$14.50.

(a) Find the price per kilogram of the 5 kg bag. (1 mark)

(b) Which bag is the better buy? (1 mark)

(c) How much is saved by buying one 5 kg bag instead of five 1 kg bags? (1 mark)

(d) A recipe uses 150 g of rice per serve. How many full serves can be made from the 5 kg bag? (2 marks)

3 Equally likely outcomes

ANCHOR QUESTION

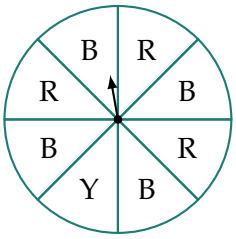
A standard die is rolled. Find P(an even number).

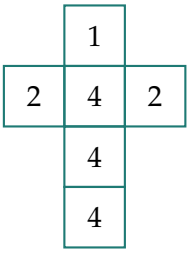
SYLLABUS MST-12-S1-07

- Express the probability of an event that has a finite number of equally likely outcomes, as $P(\text{event}) = \dfrac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$

NOTES

- Outcomes are **equally likely** when each one has the same chance of happening, such as the faces of a fair die.
- When the outcomes are equally likely, $P(\text{event}) = \dfrac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$.
- P(even) is read "the probability of an even number". Simplify the fraction, or write it as a decimal or percentage if asked.
- On a spinner, count the **equal sections**, not the colours: 3 red sections out of 8 gives $P(\text{red}) = \frac{3}{8}$.
- Repeated letters or faces count every time they appear.





WORKED EXAMPLE

A standard die is rolled. Find P(a number greater than 4).

Favourable outcomes: 5, 6 — list the outcomes that fit

$P = \dfrac{2}{6}$ — favourable ÷ total

$= \dfrac{1}{3}$ — simplify

Tier 1 · Dice and coins

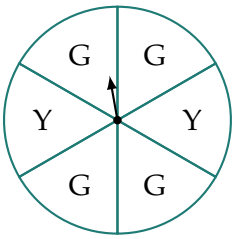
48. A fair coin is tossed. Find P(head).

49. A standard die is rolled. Find P(an odd number).

50. A standard die is rolled. Find P(a number less than 5).

Tier 2 · Spinners with repeated sections

51. For this spinner, find $P(\text{green})$.



52. A spinner has 10 equal sections: 5 red, 3 blue and 2 white. Find $P(\text{blue})$.

53. A spinner has 12 equal sections: 6 are numbered 1, 4 are numbered 2 and 2 are numbered 3. Find $P(3)$.

Tier 3 · Finding the total first

54. A bag holds 4 red and 6 green counters. One is taken out at random. Find $P(\text{red})$.

55. A jar holds 8 strawberry, 5 lime and 7 orange lollies. One is chosen at random. Find $P(\text{lime})$.

56. A class has 12 boys and 18 girls. One student is chosen at random. Find the probability that a girl is chosen, as a percentage.

Tier 4 · Letters in a word

57. A letter is chosen at random from the word BANANA. Find $P(\text{A})$.

58. A letter is chosen at random from the word STATISTICS. Find $P(\text{T})$.

59. A letter is chosen at random from the word MATHEMATICS. Find $P(\text{a vowel})$.

Tier 5 · Two conditions

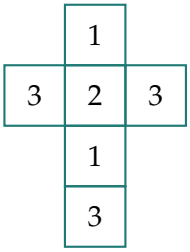
60. A number from 1 to 10 is chosen at random. Find $P(\text{an even number greater than } 5)$.

61. A number from 1 to 30 is chosen at random. Find the probability that it is a multiple of 4 or a multiple of 6.

62. A number from 1 to 50 is chosen at random. Find the probability that it is an odd multiple of 5, as a decimal.

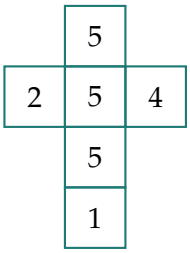
Tier 6 · Nets and biased dice

63. A die is made from this net and rolled once. Find $P(3)$.



64. A die has faces 1, 1, 1, 1, 2, 6. It is rolled once. Find $P(1)$.

65. A die is made from this net and rolled once. Which is more likely: rolling a 5 or rolling an even number?



Conceptual

66. A spinner has a red half, a blue quarter and a green quarter. Lily says $P(\text{red}) = \frac{1}{3}$ because there are three colours. Explain her mistake and find the correct probability.

67. A footy match can end in a win, a draw or a loss. Explain why $P(\text{draw})$ is not $\frac{1}{3}$.

68. A bag holds 12 marbles, some red and the rest blue. $P(\text{red}) = \frac{1}{4}$. How many blue marbles are in the bag?

69. Design a spinner with 8 equal sections, using only red and blue, so that $P(\text{blue}) = 75\%$.

70. A wheel has the numbers 1 to 40 marked on it. When the wheel is spun, each number is equally likely to win.

(a) List all the multiples of 6 on the wheel. (1 mark)

(b) Find the probability that the winning number is a multiple of 6. Give your answer in simplest form. (1 mark)

(c) Find the probability that the winning number is a multiple of 6 that is greater than 20, as a percentage. (2 marks)

Revision

71. A letter is chosen at random from the word TOMORROW.

(a) List the sample space. (1 mark)

(b) How many outcomes are in the sample space? (1 mark)

(c) Find $P(\text{O})$ as a fraction, a decimal and a percentage. (2 marks)

(d) Describe this chance in words. (1 mark)

72. A map has a scale of $1 : 25\,000$.

(a) Two landmarks are 8 cm apart on the map. Find the actual distance in km. (2 marks)

(b) A walking track is 3.5 km long. Find its length on the map in cm. (2 marks)

(c) Kai walks the track in 50 minutes. Find his average speed in km/h. (1 mark)

73. A phone plan costs \$45 per month. GST is included in the price.

(a) Find the cost of the plan for one year. (1 mark)

(b) The price rises by 8%. Find the new monthly price. (2 marks)

(c) Find the GST in the new monthly price, to the nearest cent. (2 marks)

4 Complementary events

ANCHOR QUESTION

The probability that a bus is late is 0.2. *Find the probability that the bus is not late.*

SYLLABUS MST-12-S1-07

- Apply $P(\text{an event does not occur}) = 1 - P(\text{the event does occur})$ to calculate the probability of the complement of an event

NOTES

- The **complement** of an event is the event **not** happening: the complement of rolling a 6 is rolling 1, 2, 3, 4 or 5.
- An event and its complement cover every outcome, so their probabilities add to 1.
- $P(\text{an event does not occur}) = 1 - P(\text{the event does occur})$.
- With percentages, subtract from 100%: $P(\text{rain}) = 30\%$ gives $P(\text{no rain}) = 70\%$.
- If there are several outcomes, all their probabilities add to 1.

WORKED EXAMPLE

The probability that a train is cancelled is 0.06. *Find the probability that the train is not cancelled.*

$$\begin{aligned} P(\text{not cancelled}) &= 1 - P(\text{cancelled}) && \text{complement} \\ &= 1 - 0.06 && \text{substitute} \\ &= 0.94 \end{aligned}$$

Tier 1 · Complement of a decimal

74. Describe the complement of each event.

(a) A coin lands on tails.

(b) A die shows a number greater than 4.

75. The probability that a flight is delayed is 0.15. Find the probability that it is not delayed.

76. The probability that Ana scores a goal in a match is 0.42. Find the probability that she does not score.

Tier 2 · Fractions

77. The probability of winning a prize is $\frac{1}{12}$. Find the probability of not winning.

78. The probability that a seed sprouts is $\frac{7}{9}$. Find the probability that it does not sprout.

79. The probability that a randomly chosen person is left-handed is $\frac{3}{25}$. Find the probability that the person is not left-handed.

Tier 3 · Percentages

80. A forecast gives a 65% chance of rain. Find the chance of no rain.

81. 2.5% of eggs from a farm are cracked. Find the probability that a randomly chosen egg is not cracked.

82. A basketballer makes 78% of her free throws. Find the probability that she misses her next free throw, as a decimal.

Tier 4 · Find the probability first

83. A standard die is rolled. Find $P(\text{not a } 2)$.

84. A bag holds 6 red, 9 blue and 5 yellow marbles. One is taken out at random. Find the probability that it is not yellow.

85. A letter is chosen at random from the word ELEPHANT. Find the probability that it is not an E.

Tier 5 · Several outcomes

86. A game ends in a win, a draw or a loss. $P(\text{win}) = 0.5$ and $P(\text{draw}) = 0.1$. Find $P(\text{loss})$.

87. A traffic light is green 45% of the time and amber 5% of the time. Otherwise it is red. Find the probability that it is red when a car arrives.

88. Students travel to school by car, bus or walking. $P(\text{car}) = \frac{1}{4}$ and $P(\text{bus}) = \frac{2}{5}$. Find $P(\text{walking})$.

Conceptual

89. Explain why the probability of an event and the probability of its complement must add to 1.

90. Ben says that if $P(\text{rain}) = 0.3$, then $P(\text{no rain}) = 0.3$ too. Explain his mistake.

91. The probability that a lamp is *not* faulty is 0.97. Find the probability that it is faulty.

92. A spinner has 4 colours. $P(\text{red}) = 0.3$, $P(\text{blue}) = 0.25$ and $P(\text{green}) = 0.4$. Can these probabilities be correct? Explain.

93. A wheel has the numbers 0 to 29 marked on it. When the wheel is spun, each number is equally likely to win.

(a) List all the odd numbers on the wheel that are greater than 18. (1 mark)

(b) What is the probability that the winning number is NOT an odd number greater than 18? (2 marks)

Revision

94. A bag holds 3 red, 5 white and 4 black counters. One counter is taken out at random.

(a) List the sample space. (1 mark)

(b) Find $P(\text{black})$ in simplest form. (1 mark)

(c) Find $P(\text{red})$ as a percentage. (1 mark)

(d) Is drawing a white counter likely or unlikely? Justify your answer. (2 marks)

95. A car travels 480 km and uses 38.4 L of fuel.

(a) Find the fuel consumption in L/100 km. (1 mark)

(b) Fuel costs \$1.89 per litre. Find the fuel cost, to the nearest cent. (1 mark)

(c) The trip took 5 hours 20 minutes. Find the average speed in km/h. (2 marks)

(d) How much fuel would the car use to travel 650 km? (1 mark)

96. Leo buys a laptop priced at \$1240 on a buy now, pay later scheme. He pays 4 equal fortnightly payments and a \$12 fee.

(a) Find the amount of each fortnightly payment, before the fee. (1 mark)

(b) Find the total Leo pays. (1 mark)

(c) The store offers 5% off for paying cash upfront. How much would Leo save by paying cash instead of using the scheme? (3 marks)

5 Two-stage events: arrays

ANCHOR QUESTION

A coin is tossed and a die is rolled. Draw an array and find $P(\text{head and an even number})$.

SYLLABUS MST-12-S1-07

- Represent equally likely outcomes using arrays and tree diagrams to solve problems, involving replacement where applicable, in two-stage events

NOTES

- A **two-stage event** has two parts, such as tossing a coin and then rolling a die.
- An **array** is a table of every pair of outcomes: the first stage down the side, the second stage across the top.
- Number of outcomes = rows × columns. A coin and a die give $2 \times 6 = 12$ outcomes.
- When each stage is fair, every cell is equally likely: $P = \dfrac{\text{cells that fit}}{\text{total cells}}$.
- **Without replacement**, the same item cannot be chosen twice, so cross out the cells where both stages are the same.

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Totals when two dice are rolled: 36 equally likely outcomes

WORKED EXAMPLE

Two dice are rolled and the numbers are added. Find $P(\text{total of } 9)$.

Draw the array of totals (above). — 36 cells

Totals of 9: $(3,6), (4,5), (5,4), (6,3)$ — count the cells that fit

$P = \dfrac{4}{36} = \dfrac{1}{9}$ — cells that fit ÷ total cells

Tier 1 · Counting the outcomes

97. Two coins are tossed. Draw an array and list all the outcomes. How many are there?

98. A coin is tossed and a spinner with sections red, blue and green is spun. How many outcomes are there?

99. A die is rolled and a spinner with 5 equal sections is spun. How many outcomes are there?

Tier 2 · One pair of outcomes

100. Two coins are tossed. Find $P(\text{two heads})$.

101. A coin is tossed and a die is rolled. Find $P(\text{tail and } 3)$.

102. A coin is tossed and a die is rolled. Find $P(\text{head and a number greater than } 4)$.

Tier 3 · Totals of two dice

103. Two dice are rolled and the numbers are added. Find $P(\text{total of } 8)$.

104. Two dice are rolled and the numbers are added. Find the probability that the total is less than 5.

105. Two dice are rolled. Find the probability of rolling a double (both dice the same).

Tier 4 · Products and differences

106. Two dice are rolled and the numbers are multiplied. Find $P(\text{product of } 12)$.

107. Two dice are rolled and the smaller number is subtracted from the larger. Find the probability that the difference is 2.

108. Two spinners each have 3 equal sections numbered 1, 2 and 3. Both are spun and the numbers are multiplied. Find the probability that the product is odd.

Tier 5 · With and without replacement

109. Three cards numbered 1, 2 and 3 are shuffled. Two are drawn, one after the other, without replacement. How many outcomes are there? Find the probability that the two numbers add to 4.

110. A bag holds 5 counters numbered 1 to 5. Two are drawn, one after the other, without replacement. Find the probability that both numbers are odd.

111. Repeat the last question, but the first counter is put back (with replacement) before the second is drawn.

Tier 6 · Complements in arrays

112. Two dice are rolled and the numbers are added. Find the probability that the total is not 7.

113. Two dice are rolled. Find the probability of rolling at least one 6.

114. Two coins are tossed. Find the probability of not getting two tails.

Conceptual

115. The totals for two dice are 2, 3, ..., 12. Explain why $P(\text{total of } 2)$ is not $\frac{1}{11}$.

116. Which total is most likely when two dice are rolled? Explain using the array.

117. Explain how the array changes when two items are chosen without replacement instead of with replacement.

118. A die is rolled and a spinner is spun. The array has 24 cells. How many equal sections does the spinner have?

119. Spinner A has 3 equal sections numbered 1, 2, 3. Spinner B has 4 equal sections numbered 1, 2, 3, 4. Both are spun and the numbers are multiplied.

(a) Draw an array to show all the possible products. (2 marks)

(b) Find the probability that the product is 6. (1 mark)

(c) Find the probability that the product is greater than 4. (1 mark)

Revision

120. A raffle sells 400 tickets. Zoe buys 8 tickets.

(a) Find the probability that Zoe wins, in simplest form. (1 mark)

(b) Write this probability as a percentage. (1 mark)

(c) Find the probability that Zoe does not win, as a decimal. (1 mark)

(d) How many more tickets give Zoe a 5% chance of winning? (2 marks)

121. A rectangular garden is 12.5 m long and 8 m wide.

(a) Find the area of the garden. (1 mark)

(b) Find the perimeter of the garden. (1 mark)

(c) Fencing costs \$34.50 per metre. Find the cost to fence the garden. (1 mark)

(d) Turf costs \$18.90 per m^2. Find the cost to turf the whole garden. (2 marks)

122. Trent and Sasha share \$360 in the ratio 4 : 5.

(a) Find each person's share. (2 marks)

(b) Find Sasha's share as a percentage of the total, to 1 d.p. (1 mark)

(c) Sasha earned her share at \$25 per hour. How many hours did she work? (1 mark)

(d) Trent saves \$20 a week. How many weeks to save his share? (1 mark)

6 Two-stage events: tree diagrams

ANCHOR QUESTION

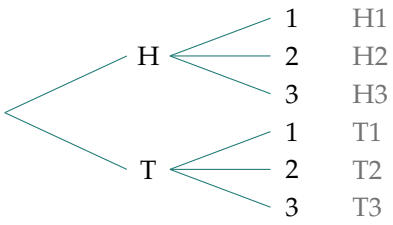
A coin is tossed, then a spinner numbered 1, 2 and 3 is spun. Draw a tree diagram and find the probability of a head and an odd number.

SYLLABUS MST-12-S1-07

- Represent equally likely outcomes using arrays and tree diagrams to solve problems, involving replacement where applicable, in two-stage events

NOTES

- A **tree diagram** shows a two-stage event as branches: the first stage, then the second stage from the end of each branch.
- Each path from start to end is **one outcome**. List the outcomes at the ends of the paths.
- When every branch at each stage is equally likely, every path is equally likely, so $P =$ paths that fit $\div$ total paths.
- **With replacement**, every second-stage branch shows all the options. **Without replacement**, the item chosen first is left off its second-stage branches.
- "At least one" means one or more.



WORKED EXAMPLE

Two coins are tossed. Draw a tree diagram and find P(at least one tail).

Paths: HH, HT, TH, TT — 4 equally likely paths
At least one tail: HT, TH, TT — one or two tails
$P = \frac{3}{4}$ — paths that fit $\div$ total paths

Tier 1 · Drawing the tree

123. A coin is tossed, then a spinner with red, blue and green sections is spun. Draw a tree diagram and list the outcomes.

124. A drink (juice or water) and a fruit (apple, banana or orange) are chosen. How many outcomes does the tree diagram show?

125. Two spinners each have 4 equal sections labelled A, B, C, D. How many paths does the tree diagram have?

Tier 2 · One path

126. Two coins are tossed. Find $P(\text{two tails})$.

127. A coin is tossed, then a spinner with red, blue and green sections is spun. Find $P(\text{tail and blue})$.

128. A coin is tossed, then a spinner numbered 1 to 4 is spun. Find $P(\text{head and } 4)$.

Tier 3 · Several paths

129. A coin is tossed, then a spinner numbered 1 to 4 is spun. Find $P(\text{tail and an even number})$.

130. A family has two children, each equally likely to be a boy or a girl. Find $P(\text{one boy and one girl})$.

131. A lunch special is chosen at random: a burger or a wrap, then chips, salad or rice. Find the probability that it does not include chips.

Tier 4 · At least one

132. Two coins are tossed. Find $P(\text{at least one head})$.

133. Two spinners each have equal sections coloured red and yellow. Both are spun. Find $P(\text{at least one red})$.

134. Two spinners each have equal sections red, yellow and green. Both are spun. Find $P(\text{at least one green})$.

Tier 5 · With and without replacement

135. A bag holds a red, a blue and a white counter. Two are taken out, one after the other, without replacement. Draw a tree diagram and find $P(\text{the blue counter is taken first})$.

136. Using the same bag without replacement, find the probability that the white counter is not taken out.

137. The same bag is used with replacement. Find the probability that the two counters are different colours.

Conceptual

138. Explain why a tree diagram for choosing two people from three, without replacement, has 6 paths and not 9.

139. Sam says $P(\text{one head and one tail}) = \frac{1}{3}$ when two coins are tossed, because the results are 2 heads, 2 tails, or 1 of each. Explain his mistake.

140. Explain how "at least one head" can be found using its complement.

141. A tree diagram has 3 first branches and 12 paths. With replacement, how many options are there at the second stage?

142. A president and then a secretary are chosen at random from Ali, Bree, Cam and Dee. The same person cannot be chosen twice.

(a) Draw a tree diagram to show all the possible outcomes. (2 marks)

(b) Find the probability that Bree is president. (1 mark)

(c) Find the probability that Ali is not chosen for either role. (2 marks)

Revision

143. Two dice are rolled and the numbers are added.

(a) How many outcomes are there? (1 mark)

(b) Find $P(\text{total of } 6)$. (1 mark)

(c) Find $P(\text{total of 11 or more})$. (1 mark)

(d) Find the probability that the total is not 11 or more. (2 marks)

144. Jess is paid \$31.20 per hour. She is paid time-and-a-half on Saturdays and double time on Sundays.

(a) Find her Saturday rate of pay. (1 mark)

(b) Find her Sunday rate of pay. (1 mark)

(c) One week Jess works 30 hours on weekdays, 6 hours on Saturday and 4 hours on Sunday. Find her pay for the week. (3 marks)

145. A runner completes a 10 km race in 48 minutes.

(a) Find her average speed in km/h. (2 marks)

(b) Convert this speed to m/s, correct to one decimal place. (1 mark)

(c) At the same speed, how long would she take to run 21.1 km? Give your answer in hours and minutes, to the nearest minute. (2 marks)

7 Relative frequency

ANCHOR QUESTION

A die is rolled 50 times and shows a 4 ten times. Find the relative frequency of rolling a 4.

SYLLABUS MST-12-S1-07

- Use relative frequency as an estimate of probability
- Recognise that increasing the number of trials produces relative frequencies that become closer in value to the theoretical probability

NOTES

- A **trial** is one go of an experiment, such as one toss of a coin.
- Relative frequency $= \dfrac{\text{number of times the event happens}}{\text{total number of trials}}$.
- Relative frequency is an **estimate** of the probability. It is used when the outcomes are not equally likely, or when the probability cannot be worked out.
- The **theoretical probability** is worked out from equally likely outcomes, such as $P(6) = \frac{1}{6}$ for a fair die.
- The more trials, the closer the relative frequency usually gets to the theoretical probability.

WORKED EXAMPLE

A spinner is spun 80 times: red 30, blue 34, green 16. Find the relative frequency of blue.

$\text{Total trials} = 30 + 34 + 16 = 80$ — add the frequencies

$\text{Relative frequency} = \dfrac{34}{80} = 0.425$ — times blue ÷ total

Tier 1 · One event

146. A coin is tossed 40 times and lands on tails 22 times. Find the relative frequency of tails.

147. In a survey of 250 people, 85 said they cycle to work. Find the relative frequency of cycling to work.

148. A goalkeeper saves 18 of 60 penalty shots. Find the relative frequency of a save, as a percentage.

Tier 2 · From a frequency table

149. A spinner is spun 60 times.

Colour	Red	Blue	Yellow
Frequency	21	27	12

Find the relative frequency of yellow.

150. The number of people in 40 cars passing a school was recorded.

People in car	1	2	3	4
Frequency	18	12	6	4

Find the relative frequency of a car with 2 people.

151. Using the car data above, find the relative frequency of a car with more than one person.

Tier 3 · Comparing with the theory

152. A coin is tossed 20 times and lands heads 13 times. Find the relative frequency of heads and compare it with the theoretical probability.

153. A die is rolled 120 times and shows a 3 twenty-one times. Find the relative frequency of a 3, to two decimal places, and compare it with the theoretical probability.

154. A spinner has 4 equal sections. In 200 spins it lands on red 49 times. Is this result close to what you would expect? Explain.

Tier 4 · More trials

155. A coin is tossed in three experiments: 10 tosses gave 3 heads, 100 tosses gave 46 heads and 1000 tosses gave 497 heads. Find each relative frequency. Which is closest to 0.5?

156. Mo rolls a die 12 times and gets no sixes. He says the die must be unfair. Explain why 12 trials is not enough to decide.

157. Two students estimate the probability that a drawing pin lands point up. Ava drops it 20 times and gets 12 point up. Ben drops it 500 times and gets 285 point up. Whose estimate is more reliable? Find both estimates.

Tier 5 · Making a decision

158. A spinner with red, white and blue sections is spun 100 times: red 51, white 25, blue 24. Which spinner is most likely? (A) three equal sections (B) red half, white quarter, blue quarter (C) red quarter, white half, blue quarter

159. A die is rolled 300 times and shows a 1 ninety times. Is the die likely to be fair? Justify your answer.

160. A factory tests 800 batteries and finds 12 faulty. Estimate the probability that a battery from the factory is faulty, as a percentage.

Conceptual

161. Explain the difference between relative frequency and theoretical probability.

162. Explain why relative frequency is needed to estimate the probability that a drawing pin lands point up.

163. A coin is tossed 4 times and lands heads every time. Lia says the probability of heads for this coin is 1. Explain her mistake.

164. The relative frequency of an event after 50 trials is 0.4. How many times did the event happen?

165. A spinner has 5 equal sections numbered 1 to 5. It is spun 150 times.

Number	1	2	3	4	5
Frequency	28	33	31	27	31

(a) Find the relative frequency of the number 2. (1 mark)

(b) Find the relative frequency of an odd number. (2 marks)

(c) Compare part (b) with the theoretical probability of an odd number. (1 mark)

Revision

166. Two counters are taken, one after the other, without replacement from a bag holding counters labelled A, B, C and E.

(a) How many outcomes does the tree diagram show? (1 mark)

(b) Find $P(\text{A is taken first})$. (1 mark)

(c) Find $P(\text{both counters are vowels})$. (1 mark)

(d) Find the probability that neither counter is a vowel. (2 marks)

167. A cylindrical water tank has a radius of 1.2 m and a height of 2 m.

(a) Find the volume of the tank in m^3, to two decimal places. (2 marks)

(b) Find the capacity of the tank in litres, to the nearest litre. (1 mark)

(c) Water flows in at 15 L/min. How long does it take to fill the empty tank, to the nearest hour? (2 marks)

168. A \$900 phone is bought for a 10% deposit and 12 monthly payments of \$72.

(a) Find the deposit. (1 mark)

(b) Find the total cost of buying the phone on these terms. (2 marks)

(c) Find the extra cost as a percentage of the cash price. (2 marks)

Solutions

1. 0.4, 75% and $\frac{2}{3}$. The others are less than 0 or more than 1.

2. (a) Impossible. (b) Even chance. (c) Likely. (d) Certain.

3. $0.9 \times 100 = 90\%$. Likely.

4. $0.02 \times 100 = 2\%$. Unlikely.

5. $2 \div 5 = 0.4 = 40\%$. Unlikely.

6. $17 \div 20 = 0.85 = 85\%$. Likely.

7. $1 \div 8 = 0.125 = 12.5\%$. Unlikely.

8. $\frac{60}{100} = \frac{3}{5}$, 0.6. Likely.

9. $\frac{4}{100} = \frac{1}{25}$, 0.04. Unlikely.

10. $\frac{72}{100} = \frac{18}{25}$, 0.72. Likely.

11. 0.3, 0.35, 0.4, so 0.3, 35%, $\frac{2}{5}$.

12. $\frac{7}{8} = 0.875$. In order: 0.85, $\frac{7}{8}$, 88%.

13. Lions: $9 \div 20 = 0.45$. Owls: 0.44. The Owls are least likely. The Hawks and Lions are equally likely (0.45 each).

14. (a) Even chance. (b) Impossible. (c) Certain.

15. C. Ella is 1 of 30 students, so it could happen but the chance is small.

16. Likely. Most of the counters (9 out of 10) are green.

17. A probability is between 0 and 1. 1.4 is more than 1, which would be more likely than certain.

18. $0.08 = 8\%$ and $0.5 = 50\%$. Since $8\% < 50\%$, 0.5 is more likely. He compared the digits, not the values.

19. $\frac{1}{2} = 0.5 = 50\%$

20. 0% means rain is impossible. A forecast is only an estimate, often rounded, so rain can still happen.

21. (a) $0.03 \times 100 = 3\%$. (b) Unlikely. (c) $\frac{1}{25} = 0.04 = 4\%$. Since $4\% > 3\%$, Factory B is more likely to make a faulty phone.

22. (a) $3 + 2 = 5$ parts, so $\frac{2}{5}$. (b) One part $= 4 \div 5 = 0.8$ L. Juice $= 3 \times 0.8 = 2.4$ L. (c) $\$3.60 \div 2 = \$1.80$ per litre. $2.4 \times \$1.80 = \$4.32$.

23. (a) $48 \div 60 = 0.8$, so 3.8 h. (b) $342 \div 3.8 = 90$ km/h. (c) $90\,000 \div 3600 = 25$ m/s. (d) $7.5 \times 3.42 = 25.65$ L. $25.65 \times \$1.96 = \$50.274 \approx \$50.27$.

24. (a) $72 \div 8 = 9$. (b) $\frac{8+9}{2} = 8.5$. (c) $15 - 4 = 11$. (d) Q_1 is the median of $4, 7, 7, 8$, so $Q_1 = 7$. Q_3 is the median of $9, 10, 12, 15$, so $Q_3 = 11$. IQR $= 11 - 7 = 4$.

25. $S = \{H, T\}$, 2 outcomes.

26. $S = \{A, B, C\}$, 3 outcomes.

27. $S = \{\text{Mon, Tue, Wed, Thu, Fri, Sat, Sun}\}$, 7 outcomes.

28. $S = \{G, Y\}$, 2 outcomes.

29. $S = \{R, B, W\}$, 3 outcomes.

30. $S = \{1, 2, 3\}$, 3 outcomes.

31. $S = \{M, I, S, P\}$, 4 outcomes.

32. $S = \{B, A, L, O, N\}$, 5 outcomes.

33. $S = \{S, T, A, I, C\}$, 5 outcomes.

34. 52

35. 28

36. $499 - 100 + 1 = 400$

37. 5, 6

38. 7, 14, 21, 28

39. 13, 15, 17, 19

40. The sample space lists the possible outcomes, each once. The spinner can only land on red, green or white, so there are 3 outcomes.

41. O is listed twice. Each outcome is listed once: $S = \{B, O, K\}$, so 3 outcomes.

42. Yes. Some numbers can appear on more than one section, as long as each of 1 to 5 appears at least once and no other number appears.

43. $S = \{\text{win, draw, loss}\}$, 3 outcomes.

44. (a) 30. (b) 22, 24, 26, 28, 30. (c) $30 - 5 = 25$. *cf. 2024 Std 1 HSC Q17*

45. (a) $3 \div 40 = 0.075$. (b) 7.5%. (c) Unlikely. (d) $7.5\% < 12\%$, so a car is more likely to be a ute.

46. (a) $38 \times \$28.40 = \$1079.20$. (b) $1.5 \times \$28.40 = \$42.60$ per hour. (c) $4 \times \$42.60 = \$170.40$. Total $= \$1079.20 + \$170.40 = \$1249.60$. (d) $2 \times \$1249.60 = \$2499.20$.

47. (a) $\$14.50 \div 5 = \$2.90$ per kg. (b) The 5 kg bag (\$2.90/kg is less than \$3.20/kg). (c) $5 \times \$3.20 = \$16.00$. Saving $= \$16.00 - \$14.50 = \$1.50$. (d) $5000 \div 150 = 33.3\ldots$, so 33 full serves.

48. $\frac{1}{2}$

49. 1, 3, 5: $\frac{3}{6} = \frac{1}{2}$

50. 1, 2, 3, 4: $\frac{4}{6} = \frac{2}{3}$

51. 4 green sections out of 6: $\frac{4}{6} = \frac{2}{3}$

52. $\frac{3}{10}$

53. $\frac{2}{12} = \frac{1}{6}$

54. Total $= 10$. $\frac{4}{10} = \frac{2}{5}$

55. Total $= 20$. $\frac{5}{20} = \frac{1}{4}$

56. Total $= 30$. $\frac{18}{30} = \frac{3}{5} = 60\%$

57. 6 letters, 3 are A: $\frac{3}{6} = \frac{1}{2}$

58. 10 letters, 3 are T: $\frac{3}{10}$

59. 11 letters. Vowels: A, E, A, I, so 4. $\frac{4}{11}$

60. 6, 8, 10: $\frac{3}{10}$

61. 4, 8, 12, 16, 20, 24, 28 and 6, 18, 30 (12 and 24 already counted): 10 numbers. $\frac{10}{30} = \frac{1}{3}$

62. 5, 15, 25, 35, 45: $\frac{5}{50} = 0.1$

63. 3 faces show 3: $\frac{3}{6} = \frac{1}{2}$

64. $\frac{4}{6} = \frac{2}{3}$

65. $P(5) = \frac{3}{6}$. Even: 2, 4, so $P(\text{even}) = \frac{2}{6}$. Rolling a 5 is more likely.

66. The colours are not equally likely: red covers half the spinner. $P(\text{red}) = \frac{1}{2}$.

67. The three results are not equally likely. Draws are much rarer than wins or losses.

68. Red $= \frac{1}{4} \times 12 = 3$. Blue $= 12 - 3 = 9$.

69. 75% of $8 = 6$. Six blue sections and two red sections.

70. (a) 6, 12, 18, 24, 30, 36. (b) $\frac{6}{40} = \frac{3}{20}$. (c) 24, 30, 36: $\frac{3}{40} = 7.5\%$. *cf. 2024 Std 1 HSC Q17*

71. (a) $S = \{T, O, M, R, W\}$. (b) 5. (c) 8 letters, 3 are O: $\frac{3}{8} = 0.375 = 37.5\%$. (d) Unlikely.

72. (a) $8 \times 25\,000 = 200\,000$ cm $= 2$ km. (b) 3.5 km $= 350\,000$ cm. $350\,000 \div 25\,000 = 14$ cm. (c) $3.5 \div \frac{50}{60} = 4.2$ km/h.

73. (a) $12 \times \$45 = \$540$. (b) $\$45 \times 1.08 = \$48.60$. (c) GST $= \$48.60 \div 11 = \$4.418\ldots \approx \$4.42$.

74. (a) The coin lands on heads. (b) The die shows 1, 2, 3 or 4.

75. $1 - 0.15 = 0.85$

76. $1 - 0.42 = 0.58$

77. $1 - \frac{1}{12} = \frac{11}{12}$

78. $1 - \frac{7}{9} = \frac{2}{9}$

79. $1 - \frac{3}{25} = \frac{22}{25}$

80. $100\% - 65\% = 35\%$

81. $100\% - 2.5\% = 97.5\%$

82. $100\% - 78\% = 22\% = 0.22$

83. $1 - \frac{1}{6} = \frac{5}{6}$

84. Total $= 20$. $P(\text{yellow}) = \frac{5}{20} = \frac{1}{4}$. $\quad 1 - \frac{1}{4} = \frac{3}{4}$

85. 8 letters, 2 are E. $1 - \frac{2}{8} = \frac{6}{8} = \frac{3}{4}$

86. $1 - (0.5 + 0.1) = 0.4$

87. $100\% - (45\% + 5\%) = 50\%$

88. $\frac{1}{4} + \frac{2}{5} = \frac{13}{20}$. $\quad 1 - \frac{13}{20} = \frac{7}{20}$

89. Either the event happens or it does not, so together they cover every outcome. The probability that one of them happens is certain, which is 1.

90. The two must add to 1, so $P(\text{no rain}) = 1 - 0.3 = 0.7$.

91. $1 - 0.97 = 0.03$

92. Yes. They add to 0.95, so $P(\text{fourth colour}) = 0.05$, which is between 0 and 1.

93. (a) 19, 21, 23, 25, 27, 29. (b) 30 numbers. $P(\text{odd and} > 18) = \frac{6}{30} = \frac{1}{5}$. $P(\text{NOT}) = 1 - \frac{1}{5} = \frac{4}{5}$. *cf. 2024 Std 1 HSC Q17*

94. (a) $S = \{R, W, B\}$. (b) $\frac{4}{12} = \frac{1}{3}$. (c) $\frac{3}{12} = 25\%$. (d) $P(\text{white}) = \frac{5}{12} \approx 0.42$, which is less than 0.5, so unlikely.

95. (a) $\frac{38.4}{480} \times 100 = 8$ L/100 km. (b) $38.4 \times \$1.89 = \$72.576 \approx \$72.58$. (c) 5 h 20 min $= 5\frac{1}{3}$ h. $480 \div 5\frac{1}{3} = 90$ km/h. (d) $8 \times 6.5 = 52$ L.

96. (a) $\$1240 \div 4 = \$310$. (b) $\$1240 + \$12 = \$1252$. (c) Cash price $= \$1240 \times 0.95 = \$1178$. Saving $= \$1252 - \$1178 = \$74$.

97. HH, HT, TH, TT: 4 outcomes.

98. $2 \times 3 = 6$

99. $6 \times 5 = 30$

100. $\frac{1}{4}$

101. $\frac{1}{12}$

102. H5, H6: $\frac{2}{12} = \frac{1}{6}$

103. $(2,6), (3,5), (4,4), (5,3), (6,2)$: $\frac{5}{36}$

104. Totals 2, 3, 4: $1 + 2 + 3 = 6$ cells. $\frac{6}{36} = \frac{1}{6}$

105. 6 doubles: $\frac{6}{36} = \frac{1}{6}$

106. $(2,6), (3,4), (4,3), (6,2)$: $\frac{4}{36} = \frac{1}{9}$

107. $(1,3), (2,4), (3,5), (4,6)$ and the reverses: 8 cells. $\frac{8}{36} = \frac{2}{9}$

108. $(1,1), (1,3), (3,1), (3,3)$: $\frac{4}{9}$

109. $9 - 3 = 6$ outcomes. $(1,3), (3,1)$: $\frac{2}{6} = \frac{1}{3}$

110. $25 - 5 = 20$ outcomes. Both odd from 1, 3, 5 with no repeats: $3 \times 2 = 6$. $\frac{6}{20} = \frac{3}{10}$

111. 25 outcomes. Both odd: $3 \times 3 = 9$. $\frac{9}{25}$

112. $1 - \frac{6}{36} = \frac{30}{36} = \frac{5}{6}$

113. No 6: $5 \times 5 = 25$ cells. $1 - \frac{25}{36} = \frac{11}{36}$

114. $1 - \frac{1}{4} = \frac{3}{4}$

115. The 11 totals are not equally likely. There are 36 equally likely cells and only one, $(1,1)$, gives 2, so $P = \frac{1}{36}$.

116. 7. It appears in 6 cells, more than any other total.

117. The cells where the first and second items are the same are crossed out, because an item cannot be chosen twice. So there are fewer outcomes.

118. $24 \div 6 = 4$ sections.

119. (a) Rows A $= 1, 2, 3$: 1, 2, 3, 4 — 2, 4, 6, 8 — 3, 6, 9, 12. (b) $(2,3), (3,2)$: $\frac{2}{12} = \frac{1}{6}$. (c) 6, 8, 6, 9, 12: $\frac{5}{12}$.

120. (a) $\frac{8}{400} = \frac{1}{50}$. (b) 2%. (c) $1 - 0.02 = 0.98$. (d) 5% of $400 = 20$ tickets. $20 - 8 = 12$ more.

121. (a) $12.5 \times 8 = 100$ m^2. (b) $2 \times (12.5 + 8) = 41$ m. (c) $41 \times \$34.50 = \$1414.50$. (d) $100 \times \$18.90 = \$1890$.

122. (a) 9 parts, one part = \$40. Trent \$160, Sasha \$200. (b) $\frac{200}{360} \times 100 = 55.6\%$. (c) $200 \div 25 = 8$ hours. (d) $160 \div 20 = 8$ weeks.

123. HR, HB, HG, TR, TB, TG: 6 outcomes.

124. $2 \times 3 = 6$

125. $4 \times 4 = 16$

126. $\frac{1}{4}$

127. $\frac{1}{6}$

128. $\frac{1}{8}$

129. T2, T4: $\frac{2}{8} = \frac{1}{4}$

130. BG, GB: $\frac{2}{4} = \frac{1}{2}$

131. 6 paths, 4 without chips: $\frac{4}{6} = \frac{2}{3}$

132. HH, HT, TH: $\frac{3}{4}$

133. RR, RY, YR: $\frac{3}{4}$

134. 9 paths. No green: $2 \times 2 = 4$. $1 - \frac{4}{9} = \frac{5}{9}$

135. RB, RW, BR, BW, WR, WB. $\frac{2}{6} = \frac{1}{3}$

136. RB, BR: $\frac{2}{6} = \frac{1}{3}$

137. 9 paths, 3 the same. $\frac{6}{9} = \frac{2}{3}$

138. The person chosen first cannot be chosen again, so each first branch has only 2 second branches: $3 \times 2 = 6$.

139. Those three results are not equally likely. The tree has 4 equally likely paths and 2 of them (HT, TH) give one of each, so $P = \frac{1}{2}$.

140. The complement is no heads (TT). $P = 1 - P(\text{TT}) = 1 - \frac{1}{4} = \frac{3}{4}$.

141. $12 \div 3 = 4$

142. (a) 4 first branches, each with the 3 other people: 12 paths. (b) 3 of 12 paths: $\frac{3}{12} = \frac{1}{4}$. (c) Paths without Ali: $3 \times 2 = 6$. $\frac{6}{12} = \frac{1}{2}$.

143. (a) 36. (b) $(1,5), (2,4), (3,3), (4,2), (5,1)$: $\frac{5}{36}$. (c) $(5,6), (6,5), (6,6)$: $\frac{3}{36} = \frac{1}{12}$. (d) $1 - \frac{1}{12} = \frac{11}{12}$.

144. (a) $1.5 \times \$31.20 = \$46.80$. (b) $2 \times \$31.20 = \$62.40$. (c) $30 \times 31.20 + 6 \times 46.80 + 4 \times 62.40 = 936 + 280.80 + 249.60 = \$1466.40$.

145. (a) 48 min = 0.8 h. $10 \div 0.8 = 12.5$ km/h. (b) $12\,500 \div 3600 = 3.47\ldots \approx 3.5$ m/s. (c) $21.1 \div 12.5 = 1.688$ h = 1 h 41.28 min $\approx$ 1 h 41 min.

146. $\frac{22}{40} = 0.55$

147. $\frac{85}{250} = 0.34$

148. $\frac{18}{60} = 0.3 = 30\%$

149. $\frac{12}{60} = 0.2$

150. $\frac{12}{40} = 0.3$

151. $12 + 6 + 4 = 22$. $\frac{22}{40} = 0.55$

152. $\frac{13}{20} = 0.65$. Theoretical $= 0.5$. The relative frequency is higher, which can happen with only 20 trials.

153. $\frac{21}{120} = 0.175 \approx 0.18$. Theoretical $= \frac{1}{6} \approx 0.17$. They are close.

154. Relative frequency $= \frac{49}{200} = 0.245$. Theoretical $= \frac{1}{4} = 0.25$. Yes, very close.

155. 0.3, 0.46, 0.497. The 1000-toss experiment is closest.

156. With few trials, results vary a lot. Getting no sixes in 12 rolls is not unusual for a fair die. Many more trials are needed.

157. Ava $\frac{12}{20} = 0.6$, Ben $\frac{285}{500} = 0.57$. Ben's is more reliable because it uses more trials.

158. Relative frequencies 0.51, 0.25, 0.24 match $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{4}$: B. *cf. 2025 Std 1 HSC Q8*

159. $\frac{90}{300} = 0.3$. A fair die gives about 0.17. With 300 trials this is too far away, so the die is likely not fair.

160. $\frac{12}{800} = 0.015 = 1.5\%$

161. Relative frequency comes from the results of an experiment. Theoretical probability is worked out from equally likely outcomes, without doing the experiment.

162. The two ways it can land are not equally likely, so the probability cannot be worked out as favourable $\div$ total. An experiment is needed.

163. 4 trials is far too few. The relative frequency is only an estimate, and it gets reliable only with many trials.

164. $0.4 \times 50 = 20$

165. (a) $\frac{33}{150} = 0.22$. (b) $28 + 31 + 31 = 90$. $\frac{90}{150} = 0.6$. (c) Theoretical $= \frac{3}{5} = 0.6$. They are the same.

166. (a) $4 \times 3 = 12$. (b) $\frac{3}{12} = \frac{1}{4}$. (c) AE, EA: $\frac{2}{12} = \frac{1}{6}$. (d) BC, CB: $\frac{2}{12} = \frac{1}{6}$.

167. (a) $\pi \times 1.2^2 \times 2 = 9.047\ldots \approx 9.05$ m^3. (b) $9.047\ldots \times 1000 \approx 9048$ L. (c) $9048 \div 15 = 603.2$ min ≈ 10.05 h, so about 10 hours.

168. (a) \$90. (b) $\$90 + 12 \times \$72 = \$90 + \$864 = \$954$. (c) $\$954 - \$900 = \$54$. $\frac{54}{900} \times 100 = 6\%$.